

6	(a)	${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\alpha$		Correct notation for alpha particle Total A number conserved and total Z number conserved	M1 A1
	(b)	(i)	Initial k.e. of the alpha particle $= 25 \times (5 \times 10^3) \times (5.2 \times 10^{-18})$ $= 6.5 \times 10^{-13} \text{ J}$ Assumption: (1) All the initial kinetic energy of the alpha particle was used to produce ion-pairs. (2) No energy was used to excite molecules that the alpha particle encountered.		M1 B1
	(b)	(ii)	1.	$\frac{1}{2} (4 \times 1.66 \times 10^{-27}) v^2 = 6.5 \times 10^{-13}$ $\therefore v = 1.399 \times 10^7 \approx 1.4 \times 10^7 \text{ m s}^{-1}$	M1 A1
	(b)	(i)	2.	By conservation of momentum, $0 = m_\alpha v_\alpha + m_{\text{Th}} v_{\text{Th}}$ $0 = (4u \times 1.399 \times 10^7) + (234u \times v_{\text{Th}})$ $v_{\text{Th}} = -2.39 \times 10^5 \approx 2.4 \times 10^5 \text{ m s}^{-1} \text{ (speed)}$	M1 A1
	(c)	(i)	$\lambda = \frac{\ln 2}{t_{1/2}}$ $\lambda = \frac{0.693}{24.1 \times 24 \times 3600}$ $= 3.328 \times 10^{-7} \approx 3.3 \times 10^{-7} \text{ s}^{-1}$		M1 A1
	(c)	(ii)	$A = \lambda N$ $A = \lambda \times \left(\frac{m}{M_r} \times N_A \right)$ $= (3.33 \times 10^{-7}) \times \left(\frac{5}{234} \times 6.02 \times 10^{23} \right)$ $= 4.28 \times 10^{15} \approx 4.3 \times 10^{15} \text{ Bq}$		M1 A1

7	(a)	(i)	1. The frequency of the source is the number of oscillations per unit time	B1
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